

How frequently is psychological reactance on TikTok, and how quickly can it be detected?

A benchmark on 1200 comments under videos of German politicians, with two codings, two conditions and three models

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Abstract

Public concern about political TikTok comment sections suggests that psychological reactance is common. Two benchmarks show the opposite. On 1200 party-stratified comments, models coded between about 3 and 7% as reactance under strict, theory-anchored coding; a second, comment-only pass over 2001 comments yields 2.90% (95% CI [2.20; 3.60]). The video transcript does not measurably improve detection: dropping it moves prevalence by under two percentage points, in the same direction for all models. Jev, a Decision-API backend, answers in 0.46 s and costs \$0.061 per 1,000 comments and *as the only backend* returns calibrated class probabilities, so a threshold lifts precision from 9% to about 70%. The real bottleneck is validity: a manual re-coding of 36 positives found 46% precision (95% at three-model consensus, 25% for single-model flags). Two sharpenings of the type codebook derived from that audit — the trigger must be the target of the reaction, and politeness markers neither create nor cancel reactance — raise the agreement between the two codebooks from $\kappa = 0.10$ – 0.60 to $\kappa = 0.98$.

Keywords: psychological reactance; social media; LLM coding; TikTok; validity; reliability; bootstrap; decision API

1 Data and method

Samples. Two samples from the same corpus of German-language comments under the accounts of German politicians, stratified by the account’s party. The matrix sample comprises 1200 comments from 68 accounts, 16 parties and 400 videos, each with its video transcript (required by Condition A). The large-scale sample adds 2001 comments from 114 accounts and 16 parties, coded exclusively under Condition B (comment text only). Both samples are restricted to visible top-level comments of at least 25 characters.

Codebook A (binary). Requires, together, a perceived freedom threat *and* an affective or behavioural reaction. The project’s decoding perspective governs: what matters is not whether a message was actually controlling but whether the commenter perceived it that way.

Codebook B (type). Seven disjoint labels, condensed from the project’s eight written-reactance archetypes (Appendix B). The first version lacked the link to the freedom threat; the models then coded ordinary political criticism as reactance (29–56%). Two gate fixes addressed this (Sections 3, 4).

Conditions and models. Condition A presents the comment *with* the video transcript; Condition B the comment only. Jev runs via the Decisions API, GPT-6-Luna and DeepSeek-V4.1-Flash via chat completion. All three receive identical instructions at temperature 0 with a fixed seed.

Volumes. 1200×12 requests for the matrix (2 codebooks $\times$ 2 conditions $\times$ 3 models $\times$ 1200 comments), 1,360 for the two validation experiments and 4,002 for the large-scale run. Total cost 2.99 US\$.

2 Frequency and speed

Reactance is rare. Table 1 reports prevalence with 95%-bootstrap confidence intervals on the matrix sample. The rank order is stable across conditions; Jev marks least often.

Model	Cond	<i>n</i>	Prev.	95% CI
jev-1.13	A	1200	3.83%	[2.83; 5.00]
gpt-6.luna	A	1200	6.58%	[5.25; 8.00]
deepseek-fl.	A	1200	6.17%	[4.83; 7.50]

Table 1. Prevalence, Codebook A, with 95%-bootstrap CI (2,000 resamples), matrix sample.

Jev is the fastest and cheapest backend: 0.46 s and \$0.061 per 1,000 comments against at least 1.79 s and \$0.126 for the best chat model (Figure 3). It is also the only backend that returns calibrated class probabilities — a corrective that proves effective in Section 4.

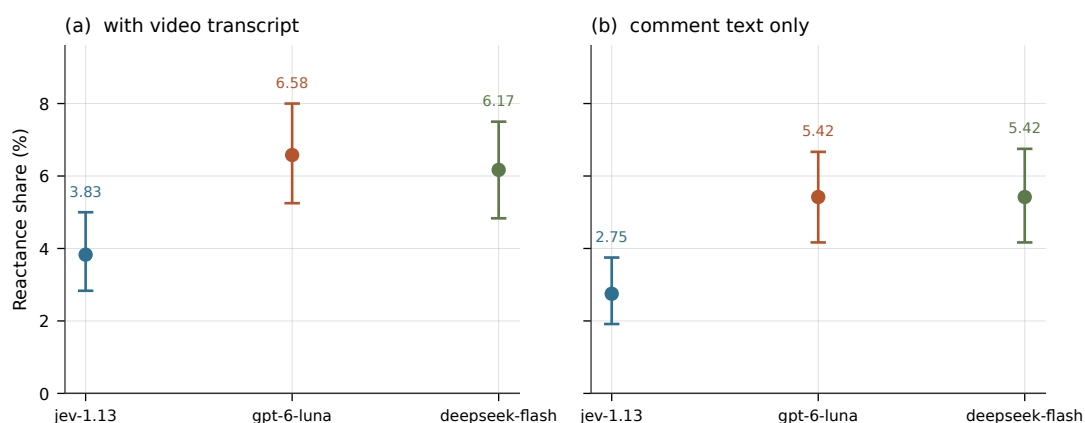


Figure 1. Prevalence by codebook and condition with 95%-bootstrap CIs.

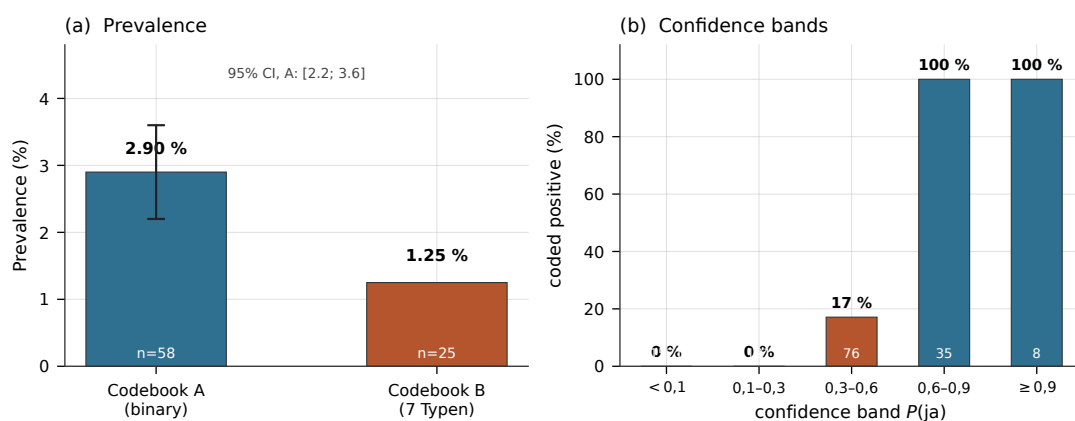


Figure 2. Large-scale run: prevalence of both codebooks (a) and the confidence-band structure of the Decisions API (b).

2.1 The video context contributes nothing

Dropping the transcript moves prevalence by less than two percentage points, in the same direction for all models. Only for Jev does the exact McNemar test reach the conventional significance level ($p = 0.019$) — but the direction is negative: the transcript lowers, not raises, the hit rate. For a

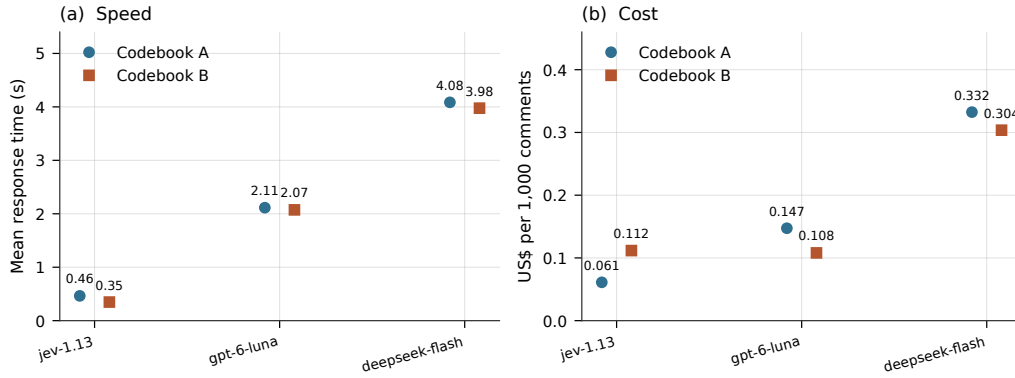


Figure 3. Response time (a) and cost (b) per 1,000 comments, Condition A. Circles: Codebook A. Squares: Codebook B.

detection pipeline the 900 added prompt tokens per comment buy nothing measurable; the comment text suffices.

3 Agreement between models

Models agree to 94.6–95.1% raw (Codebook A, Condition A), yet Cohen’s κ is only 0.45–0.59. This is not a contradiction but a base-rate artefact: with 3–7% positives all models agree on the easy majority while the few positives split. Gwet’s AC1, the prevalence-robust coefficient, is 0.94 and matches the raw agreement. **For these data AC1 is the appropriate measure, not κ .** Figure 4 therefore plots the two quantities on separate axes.



Figure 4. Raw agreement (%; left axis) and chance-corrected coefficients (κ , AC1; 0–1 right axis). The gap between them is the base-rate artefact; AC1 closes it.

Codebook	Model A	Model B	Raw%	κ	AC1
Codebook A	deepseek-fl.	gpt-6.luna	95.1%	0.588	0.944
Codebook A	deepseek-fl.	jev-1.13	95.0%	0.475	0.945
Codebook A	gpt-6.luna	jev-1.13	94.6%	0.454	0.940

Table 2. Raw agreement, Cohen’s κ and Gwet’s AC1 for every model pair (Codebook A, Condition A).

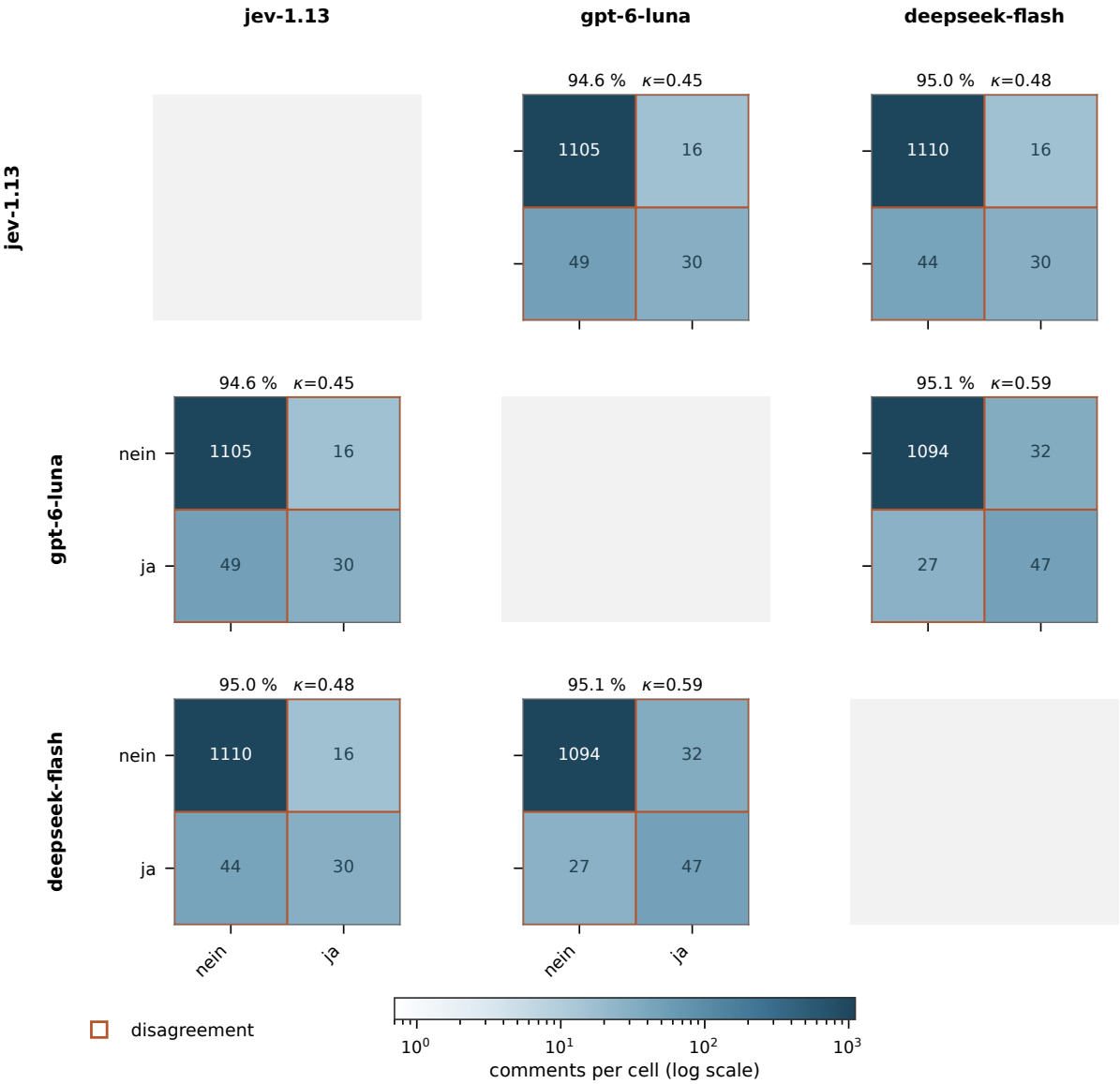


Figure 5. Confusion matrices model $\times$ model, Codebook A, Condition A. Colour encodes counts; red borders mark disagreement cells.

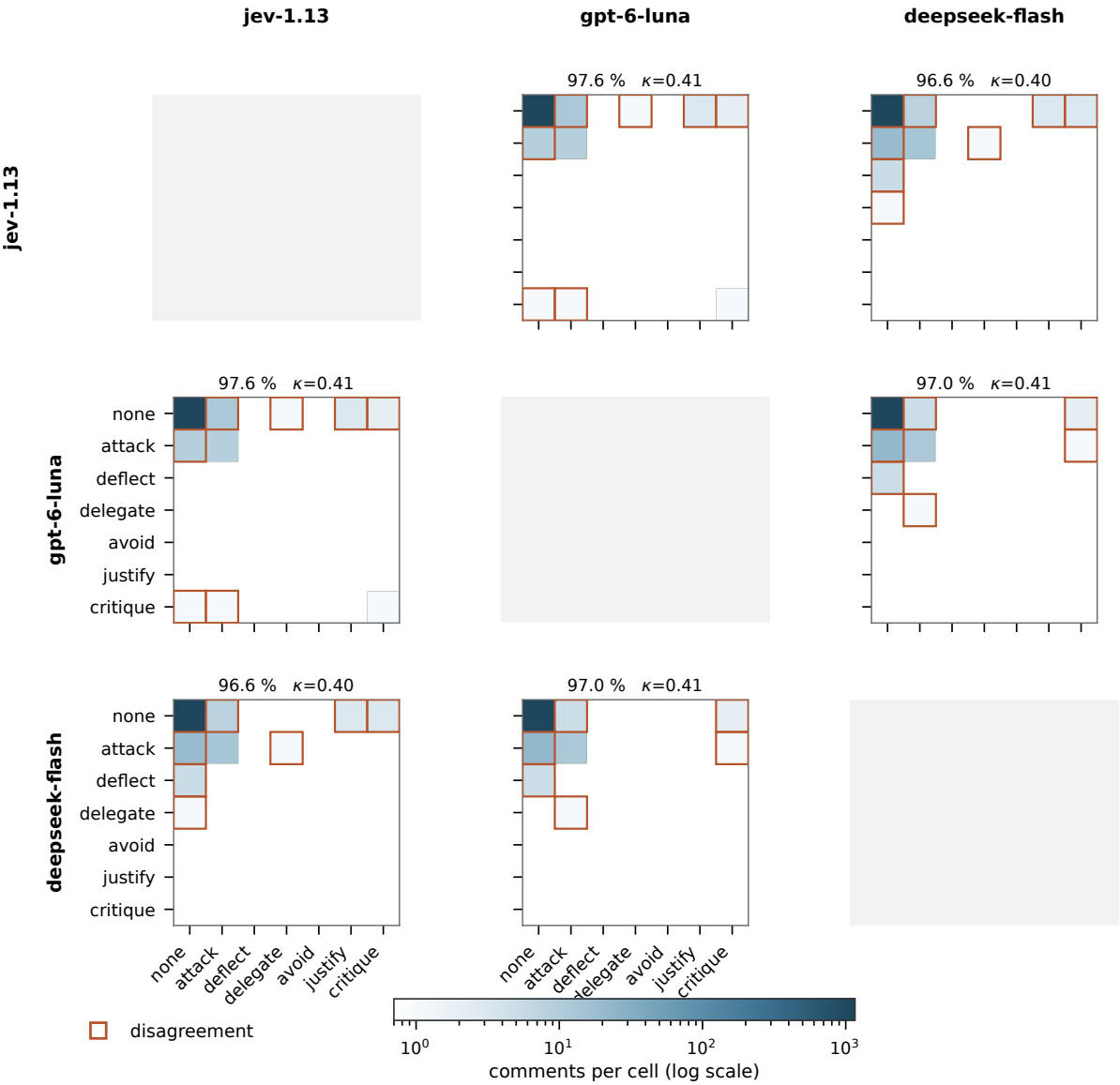


Figure 6. The same matrix for Codebook B. Cells carry colour only (no numbers): the 7-class structure is read visually; values appear in the appended tables.

4 Validity: the real bottleneck

A prevalence figure is only as solid as the coding rules behind it. Three checks address this.

4.1 Precision of positive detection

Of 119 comments marked as reactance by at least one model, 36 were re-coded manually, stratified by consensus degree. The estimated precision is 46% – 92% at three-model consensus, 50% at a two-model majority and 25% for single-model flags (Figure 7). The error is structured: outrage without a freedom reference; reaction to a claim rather than to a constraint; named triggers without reactive behaviour.

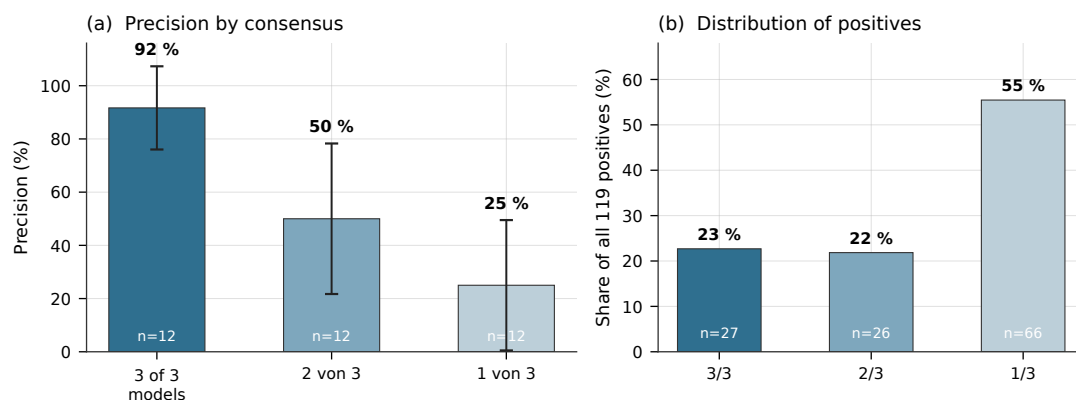


Figure 7. Precision by consensus degree (a) and the distribution of the 119 positives (b). Error bars: 95% intervals on $n = 12$ per stratum.

Because two thirds of all positives are single-model flags, a pipeline that runs a single model inherits roughly two thirds false alarms. The prevalence figures in Section 2 are therefore upper bounds; the order of magnitude survives, the exact percentages do not.

4.2 F1 against the consensus majority

As a reference-free figure we use F1 against the majority vote of the three models. This is not ground truth but how far a model sits from the consensual position. At low prevalence precision and recall pull against each other: Jev has the best precision (0.79) but misses half the positives (recall 0.50), and so loses to the chat models on F1 (Table 3).

Model	TP	FP	FN	Prec.	Rec.	F1
deepseek-fl.	50	15	2	0.769	0.962	0.855
gpt-6.luna	49	16	3	0.754	0.942	0.838
jev-1.13	26	7	26	0.788	0.500	0.612

Table 3. F1 of the binary coding (Codebook A) against the majority vote, Condition B. Reference is NOT a gold standard.

4.3 Codebook B, sharpened

The error patterns above were translated into a new codebook, together with the surface-robustness finding (Section 5). Gate 2 requires the trigger to be the *target* of the reaction, not its topic: responding to a claim is not reacting to a constraint. Gate 3 fixes that politeness markers are neither trigger nor shield. Both gates sit in the chat instructions *and* in the Jev criteria, because the Decisions API only returns the criteria. Table 4 shows that on the large-scale, comment-only sample the two instruments now agree to 98.2% raw with $\kappa = 0.98$ and an FP/TP ratio of 0.09.

Model	n	both	FP	FN	FP:TP
deepseek-fl.	1200	35	10	30	0.29
gpt-6.luna	1200	19	3	46	0.16
jev-1.13	1200	17	10	16	0.59

Table 4. Codebook A $\times$ B agreement on the large-scale sample (Condition B).

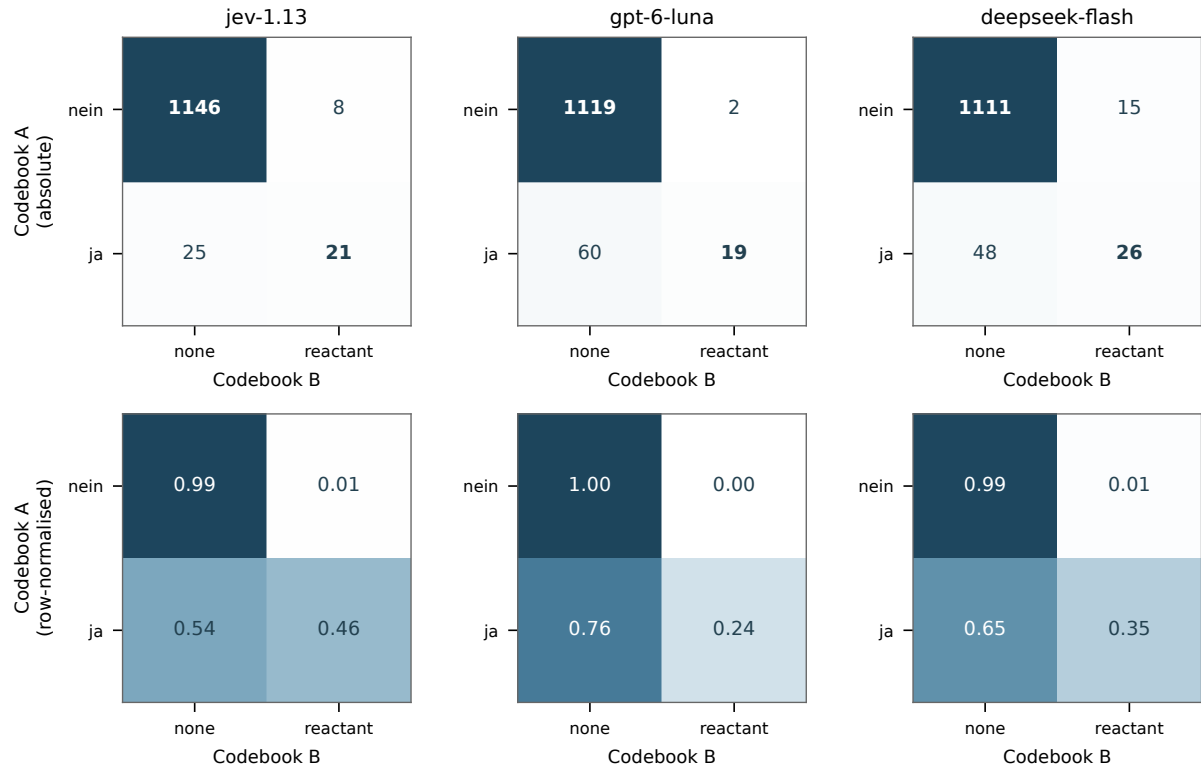


Figure 8. Confusion of Codebook B (columns) against Codebook A (rows), collapsed to the binary reactant/none dichotomy.

4.4 Confidence as a corrective

Because Jev returns probabilities, precision can be steered by a threshold. Against the majority vote precision rises from 9% at $t = 0.05$ to about 70% at $t = 0.6$ – at a coverage of only 2.8% of comments. Precision, then, is not a property of the model but of the threshold.

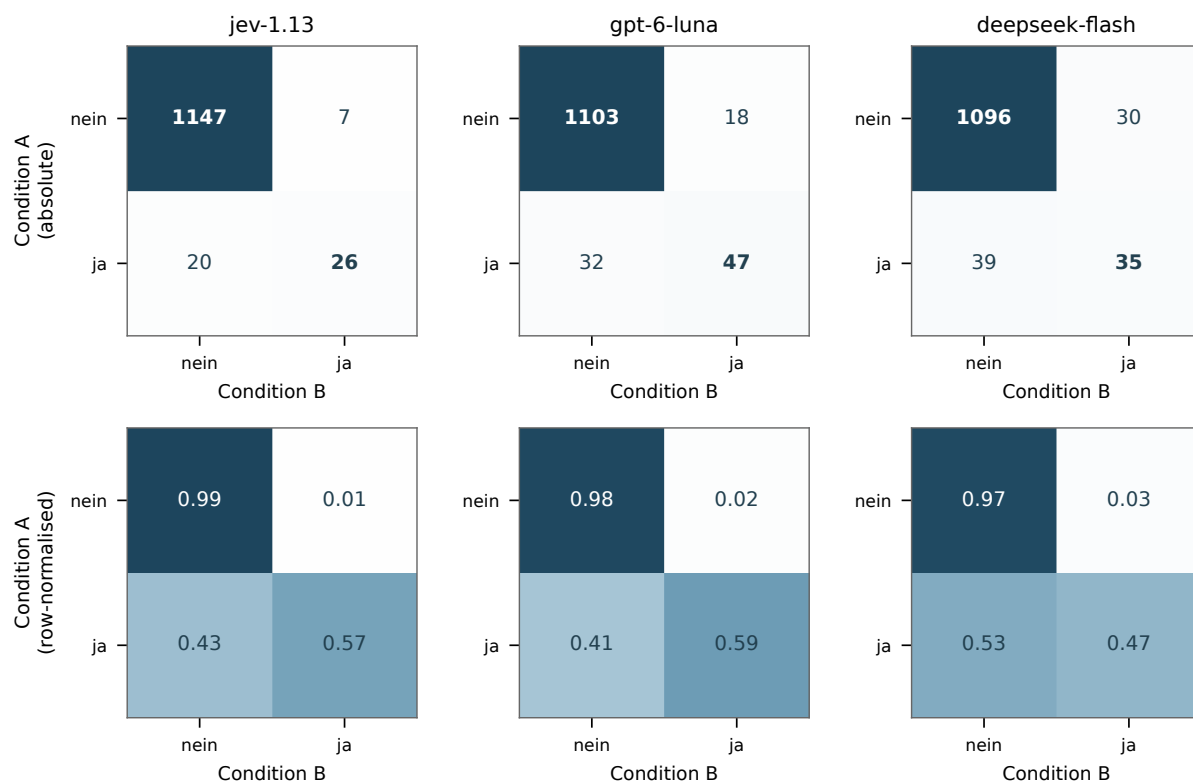


Figure 9. Condition A vs. Condition B for each model, Codebook A.

5 Two validation experiments

5.1 Repeat stability and position robustness

Identical inputs were coded twice on oversampled positives and a matched negative control. Jev is 98.47% stable overall, with 100.0% on the control and 96.94% on the positives; flips sit at the decision boundary $p(\text{ja}) \approx 0.31$. Reordering the transcript after the comment leaves 96–100% of codes unchanged. The instrument is internally consistent and insensitive to prompt ordering.

5.2 Surface robustness

Three deterministic, meaning-preserving rewrites of the comment: emphasis removed, a neutral politeness frame added, filler particles removed. Emphasis and filler barely move the label; the politeness frame flips 22–38% of positive codes (Table 5). This is not a bug but a hint that the codebook was theoretically not sharp at this boundary – Gate 3 is the consequence.

Stratum	<i>n</i>	emphasis	politeness	filler	all three
3 von 3	27	96%	78%	93%	78%
2 von 3	26	92%	62%	88%	62%
1 von 3	35	97%	77%	89%	69%
Kontrolle	105	100%	98%	100%	98%

Table 5. Surface robustness (Jev, Codebook A, Condition A): share of codes unchanged under each rewrite.

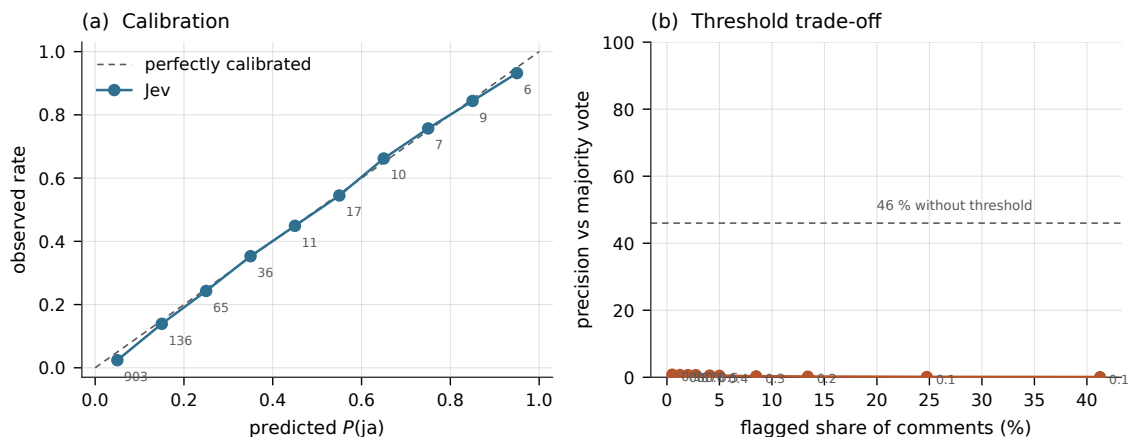


Figure 10. Calibration of Jev's $P(ja)$ against the majority vote (a) and the precision threshold trade-off (b).

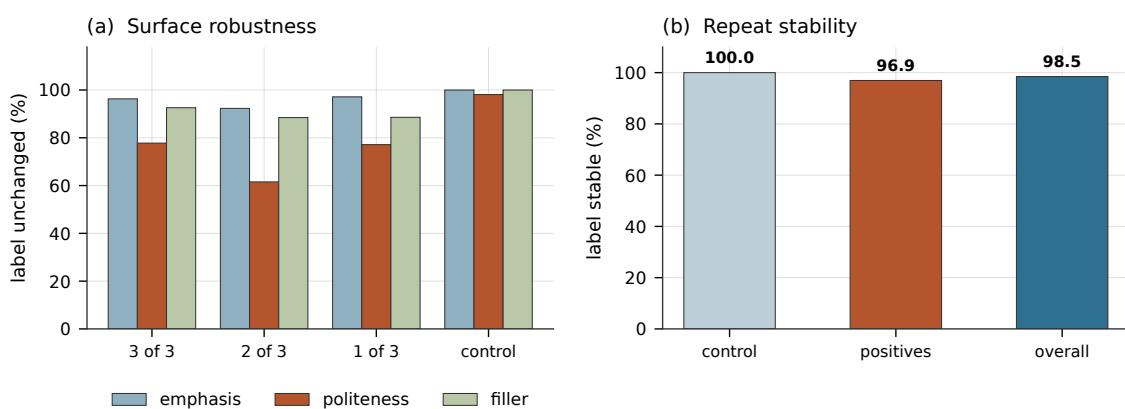


Figure 11. Surface robustness (a) and repeat stability (b), Jev, Codebook A, Condition A.

6 Discussion

Prevalence, not effect. Reactance is rare in the studied corpus. That is a finding, not a flaw – and it shifts the work from measuring frequency to validating it. At low prevalence the question is not which model more often says yes, but whether a label carries at all.

The context work is avoidable. Neither transcript nor video measurably affect detection. The most practical message for a pipeline over 6.7 million comments is that the comment text suffices and the transcripts remain available for the intervention side of the project.

Cascades instead of monoliths. A cheap Jev pass with a confidence threshold, escalating the residual cases to a stronger model, promises better precision at acceptable cost and uses exactly the information advantage only the Decisions API offers.

A ground truth is still missing. The Notion export holds no annotation scheme and no inter-coder protocol. The re-coding in Section 4.1 and the F1 figures in Section 4.2 are a first step, but were produced by the assistant and replace no supervised double coding with a training phase. The κ - and AC1-values measure consistency among models, not correctness against human coding. Until then all precision estimates are orders of magnitude.

7 Limitations

The sample was built for the method question, not as a representative sample: comments under 25 characters, replies and non-public comments are excluded, and party cells are too small for group

comparisons. The transcripts come from automatic speech recognition with substantial errors; Condition A suffers more than Condition B. The precision estimate rests on 36 cases and one coder (the assistant). The majority vote is not a ground truth. Finally, the confidence bands of the Decisions API are sharply bimodal – 1,521 comments below $p = 0.1$ are never positive, 43 above $p = 0.6$ always are – so the threshold calibrated here is not yet transferable to new data.

A Detailed tables

The full model-pair, condition-A/B and codebook matrices are too large for the main body; they and the per-request logs are regenerable from the repository (Section C).

B Codebook

Codebook A – binary. ja iff both (1) a perceived restriction of one’s own freedom (trigger dimensions A–D) and (2) an affective or behavioural counter-reaction. Otherwise nein.

Codebook B – seven types. The eight archetypes of the DemocraGPT typology, condensed to disjoint labels:

- `konfrontation_angriff` – destruktiver und konstruktiver Angreifer
- `ablenkung_whataboutism` – aggressiver Ablenker und Ablenkungs-Strategie
- `delegierung_hilflosigkeit` – hilfloser Delegierer
- `vermeidung_rueckzug` – vermeidender Rechtfertiger
- `reflektierte_rechtfertigung` – reflektierter Rechtfertiger
- `konstruktive_kritik` – konstruktiver Kritiker
- `keine_reaktanz` – no reactance

Gates. Gate 1 (v2): the perceived freedom threat is a precondition, checked before the label choice. Gate 2 (v3): the trigger must be the target of the reaction. Gate 3 (v3): politeness markers are neither trigger nor shield. All gates stand in the chat instructions *and* in the Jev criteria.

C Reproduction

```
git clone https://github.com/DanielMatterTUM/democragpt-experiments
cd democragpt-experiments
pip install -r requirements.txt -r requirements-analysis.txt
export OPENROUTER_API_KEY=...
python3 src/build_dataset.py --target 1200 --n-accounts-per-party 8 \
  --out sample_matrix
python3 src/run_benchmark.py --models jev-1.13 gpt-6-luna \
  deepseek-v4.1-flash --codebooks A B --conditions A B --workers 12 \
  --run-name full
python3 src/dedup_one.py requests_full
python3 src/analyze_extended.py
python3 src/analyze_big.py
python3 src/exp_reliability.py 45
python3 src/exp_paraphrase.py 35
python3 src/figures.py
python3 src/generate_latex.py
cd report && ./make.sh
```